

Encryption & Decryption

① Encryption

a. Caesar Cipher

Text: CHASE YOUR DREAMS

Formula: Enc: $C = E(P+n) \bmod 26$

Dec: $P = D(C-n) \bmod 26$

Key: $3 = n$

Answer: FKDVH BRXU GVHDPV

$C = 3 + 3 = 6 = F$
 $H = 8 + 3 = 11 = K$...

b. Shift Cipher

Text: CHASE YOUR DREAMS

Formula: Enc: $C = E(P+n) \bmod 26$

Dec: $P = D(C-n) \bmod 26$

Key: $7 = n$

Answer: JOHZL FVBY KYLHTZ

$C = 3 + 7 = 10 = J$
 $H = 8 + 7 = 15 = O$...

1 A	26 Z
2 B	25 Y
3 C	24 X
4 D	23 W
5 E	22 V
6 F	21 U
7 G	20 T
8 H	19 S
9 I	18 R
10 J	17 Q
11 K	16 P
12 L	15 O
13 M	14 N

C. ROT13

Text: CHASE YOUR DREAMS

Formula: Encr: $C = E(P+n) \bmod 26$

Dec: $P = E(C-n) \bmod 26$

Key: $13 = n$

Ans: PUNER LBHE QERNZF

$$C = 3 + 13 = 16 = P$$

$$H = 8 + 13 = 21 = U \dots$$

(2) Decryption

a. Caesar Cipher

Text: WRWDOOB VHFUHW PHVVDJH

Formula: $C = E(P+n) \bmod 26$

$P = D(C-n) \bmod 26$

Key: $n = 3$

Ans: TOTALLY SECRET MESSAGE

$$W = 23 - 3 = 20 = T \dots \dots$$

$$R = 18 - 3 = 15 = O$$

b. Shift Cipher

Text: SJI IBTW YNRJ

Formula: $C = E(P+n) \bmod 26$

$P = D(C-n) \bmod 26$

Key: $n = 5$

Ans: NED DWOR TIME

$$S = 19 - 5 = 14 = N$$

$$J = 10 - 5 = 5 = E$$

C. ROT13

Text: FRAQ ZBER PBSSRR

Formula: $C = E(P+n) \bmod 26$
 $P = D(P-n) \bmod 26$

Key: $n = 13$

Ans: SEND MORE COFFEE

$$F = 6 - 13 = -7 = 19 = S$$

$$R = 18 - 13 = 5 = E$$